

EcoEcon: Ecological Resilience Indicators for Systemic Risk—A Retrospective Analysis of the 2008 Financial Crisis with a 2020 Negative Control

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github.com/Abhiv1028/EcoEcon

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Abstract

Can ecological collapse theory provide early warning indicators for financial crises? We model the US economy as a biological ecosystem—sectors as species, supply chains as mutualistic dependencies, and bankruptcy cascades as extinction events—and apply three bodies of ecological theory to BEA Input-Output and FRED data: May’s (1972) stability theorem, Scheffer et al.’s (2009) critical slowing down, and network robustness metrics. We construct an Economic Biodiversity Index (EBI), track network spectral gap, and compute critical slowing down signals across two episodes: the 2008 Global Financial Crisis (endogenous fold bifurcation, signal expected) and the 2020 COVID shock (exogenous perturbation, no signal expected by theory). In a retrospective analysis, two of five ecological signals behaved as theorized before 2008: spectral gap declined monotonically 2002–2007 (Kendall $\tau = -0.87$) and housing starts variance peaked approximately 18 months before crisis onset. Logistic regression trained on these features achieves leave-one-out AUC = 1.000 ($p = 0.034$, permutation test) with zero false positives in cross-validation. The 2020 negative control holds: models trained on 2008 assign near-zero crisis probability to 2015–2019, and four of five ecological signals show no pre-crisis buildup before the exogenous COVID shock, consistent with the theoretical prediction. Spectral gap produces zero false alarms across ten calm-period years (2010–2019). This analysis is explicitly retrospective and limited to one positive crisis episode; generalizability requires multi-crisis validation. We propose ecological resilience indicators as a theoretically grounded and empirically promising direction for systemic risk research.

1 Introduction

Financial crises remain notoriously difficult to predict. Standard early warning systems rely on macroeconomic aggregates—credit growth, asset prices, current account deficits—yet the 2008 Global Financial Crisis exposed the limitations of these approaches: most official forecasts in 2007 projected continued growth (8).

Ecology faces a structurally analogous problem: detecting when a stable ecosystem is approaching collapse. Over five decades, ecologists have developed a robust theoretical framework for understanding stability and collapse in complex systems. (author?) (1) proved that complexity itself can be destabilizing—above a critical threshold of diversity and connectance, large random systems become inherently unstable. (author?) (2) demonstrated that many complex systems exhibit “critical slowing down” before tipping points: rising variance and autocorrelation in key state variables as the system loses resilience. (author?) (3) showed that financial networks exhibit phase transitions where diversification reduces individual risk up to a point, beyond which it amplifies systemic fragility through contagion channels. (author?) (4) applied this ecological analogy directly to banking systems, demonstrating that the same complexity-fragility tradeoff governs interbank contagion networks.

This paper operationalizes the ecology-economy analogy by constructing ecological metrics from economic data and testing whether they were elevated before the 2008 crisis. Critically, we also test a theoretically motivated *negative control*: the 2020 COVID shock was an exogenous perturbation rather than an endogenous approach to a bifurcation, so ecological theory predicts that critical slowing down signals should *not* appear before 2020. This contrast—signal before an endogenous crisis, no signal before an exogenous shock—provides stronger evidence than a single positive episode alone.

We make no claim that these signals would have been detectable in real time or that they generalize beyond this single positive episode. That question requires multi-crisis validation outside the scope of this retrospective study.

2 Methods

2.1 Variable Mapping

Table 1 presents the ecological-to-economic translation underlying our metrics. We note one important theoretical correction relative to standard ecological analogies: supplier-buyer relationships in IO networks are structurally closer to *mutualistic* interactions (+/+) than predator-prey interactions (+/−), since both parties benefit from the transaction. This distinction matters because (author?) (5) showed that mutualistic networks destabilize more rapidly with increasing diversity and connectance than predator-prey networks—placing economic IO networks in a potentially more fragile stability regime than the predator-prey framing would imply.

Table 1: Ecological-to-economic variable mapping

Ecological Concept	Economic Analog
Species	Sector (68 BEA industries)
Mutualistic dependency	Supplier-buyer relationship
Biomass flow	Revenue flow between sectors
Keystone species	Systemically important sector
Extinction cascade	Bankruptcy contagion
Ecosystem biodiversity	Sector centrality distribution (EBI)
Critical slowing down	Rising variance + autocorrelation
Spectral gap collapse	Network fragmentation

2.2 Data Sources

BEA Input-Output Tables. Annual sector-level flows for 68 industries from the Bureau of Economic Analysis (TableID 259), covering two episodes: 2002–2009 (2008 GFC) and 2015–2021 (2020 COVID negative control). Edge weights are normalized to technical coefficients (column-normalized input shares) rather than raw dollar flows, removing nominal GDP and inflation effects from network metrics. The 2001 dot-com episode was excluded due to incompatible industry classification systems between pre-2002 SIC-based and post-2002 NAICS-based BEA IO tables.

FRED Financial Time Series. Monthly data from the Federal Reserve Economic Database, 1998–2022: housing starts (HOUST), bank credit (TOTBKCR), TED spread (TEDRATE, substituting for BAMLH0A0HYM2 which returned empty via API), consumer credit (TOTALSL), and the Federal Funds rate (FEDFUNDS). All rolling windows are strictly trailing (no centering); detrending uses first-differencing.

Crisis Dating. NBER recession dates for both episodes. 2008 episode: December 2007 onset (NBER), June 2009 trough. 2020 episode: March 2020 onset, December 2020 trough. Episode types pre-registered before analysis: 2008 as endogenous fold bifurcation (signal expected), 2020 as exogenous shock (no signal expected).

2.3 Economic Biodiversity Index (EBI)

For each year, we construct a directed, weighted graph $G = (V, E, W)$ where nodes are the 68 BEA sectors and edge weights are technical coefficients. We compute betweenness centrality for each node and form a probability distribution:

$$p_i = \frac{c_i}{\sum_{j=1}^n c_j}, \quad i = 1, \dots, n \quad (1)$$

where c_i is the betweenness centrality of sector i . The Shannon entropy of this distribution is:

$$H = - \sum_{i=1}^n p_i \ln p_i \quad (2)$$

The Economic Biodiversity Index is:

$$\text{EBI} = H \times (1 - \text{top3_share}) \quad (3)$$

where top3_share is the fraction of total centrality held by the three most central sectors. This penalizes apparent diversity by keystone concentration. The choice of betweenness centrality and the top-3 threshold are exploratory; robustness to eigenvector centrality and DebtRank is reserved for future work.

2.4 Spectral Gap

The spectral gap is defined as:

$$\Delta\lambda = \lambda_1 - \lambda_2 \quad (4)$$

where λ_1, λ_2 are the largest and second-largest eigenvalues (by magnitude) of the technical-coefficient adjacency matrix. A collapsing spectral gap indicates network fragmentation—connectivity is becoming dominated by a single mode, reducing redundant pathways.

We note explicitly that this is a *network connectivity property* and is distinct from May’s (1972) stability criterion, which concerns the leading eigenvalue of the community (Jacobian) matrix of population dynamics near equilibrium. Computing May’s criterion properly from the IO technical-coefficient matrix requires specifying sector adjustment dynamics and self-regulation terms; this is reserved for future work.

2.5 Critical Slowing Down

Following (author?) (2), we compute two metrics on monthly FRED series using 24-month strictly trailing rolling windows:

$$\text{Variance}_t = \text{Var}(x_{t-23:t}) \quad (5)$$

$$\text{AR}(1)_t = \text{Corr}(x_{t-23:t-1}, x_{t-22:t}) \quad (6)$$

Rising variance and rising autocorrelation jointly indicate critical slowing down. This is theoretically grounded for fold bifurcations (2): as a system approaches a tipping point, the dominant eigenvalue of the linearized recovery dynamics approaches zero, causing perturbations to decay increasingly slowly. We note that CSD is only theoretically expected for *endogenous* approaches to bifurcation, not for exogenous shocks—motivating the 2020 negative control.

2.6 Trend Significance

We compute Kendall’s τ rank correlation between year and each signal over the pre-crisis window for both episodes. Kendall’s τ is appropriate for small samples with potential non-normality and is the standard in the EWS literature (2). A positive τ indicates a rising trend; the sign is adjusted so that positive always means “moving toward risk” (rising for variance signals, declining for biodiversity/connectivity signals).

2.7 Machine Learning Evaluation

Labels. For the 2008 episode, crisis labels are assigned to 2008–2009 only. The year 2007 is held out as an “advance warning test year”—it receives no label and is tested separately to assess whether models fire before the official crisis onset.

Scaler. A RobustScaler is fitted on pre-crisis calm years only (2002–2006) and applied to all subsequent data. This prevents look-ahead leakage through feature normalization.

Evaluation. Leave-one-out cross-validation on the 7-sample annual training set (2002–2006 calm + 2008–2009 crisis). We report mean LOO AUC as the headline metric, not best-seed or full-sample AUC. Statistical significance is assessed via permutation test (1000 permutations of crisis labels).

2020 negative control. Models trained on the 2008 episode are applied to the 2020 episode without retraining. Theory predicts near-zero crisis probability for 2015–2019 (pre-COVID calm).

3 Results

3.1 Signal Analysis: Which Ecological Metrics Work?

Table 2 summarizes the pre-crisis behavior of all five ecological signals across both episodes. Figure 1 shows the signal trajectories.

Table 2: Ecological signal behavior across episodes. Kendall τ computed over pre-crisis years. For the 2020 negative control, theory predicts no significant trend (i.e., $|\tau|$ should be small and p large).

Signal	2008 GFC (expect signal)		2020 COVID (expect no signal)	
	τ	Verdict	τ	Verdict
Spectral gap	-0.87	✓ Declining	+0.20	✓ No trend
Housing starts var	+1.00	✓ Rising (peaks 2006)	-0.80	✓ Declining
EBI	+0.20	✗ Wrong direction	+0.80	~ Rising (network change)
Housing starts AR(1)	-0.60	✗ Wrong direction	-0.40	✓ No buildup
TED spread var	+0.40	~ Coincident only	+1.00	✗ Slow rise (FP risk)
Summary	2/5 signals as theorized		3/5 clean, 1 ambiguous, 1 FP	

Figure 1: Ecological early warning signals — 2008 GFC episode
Red shading = crisis period (Dec 2007–Jun 2009). Dashed line = crisis onset.

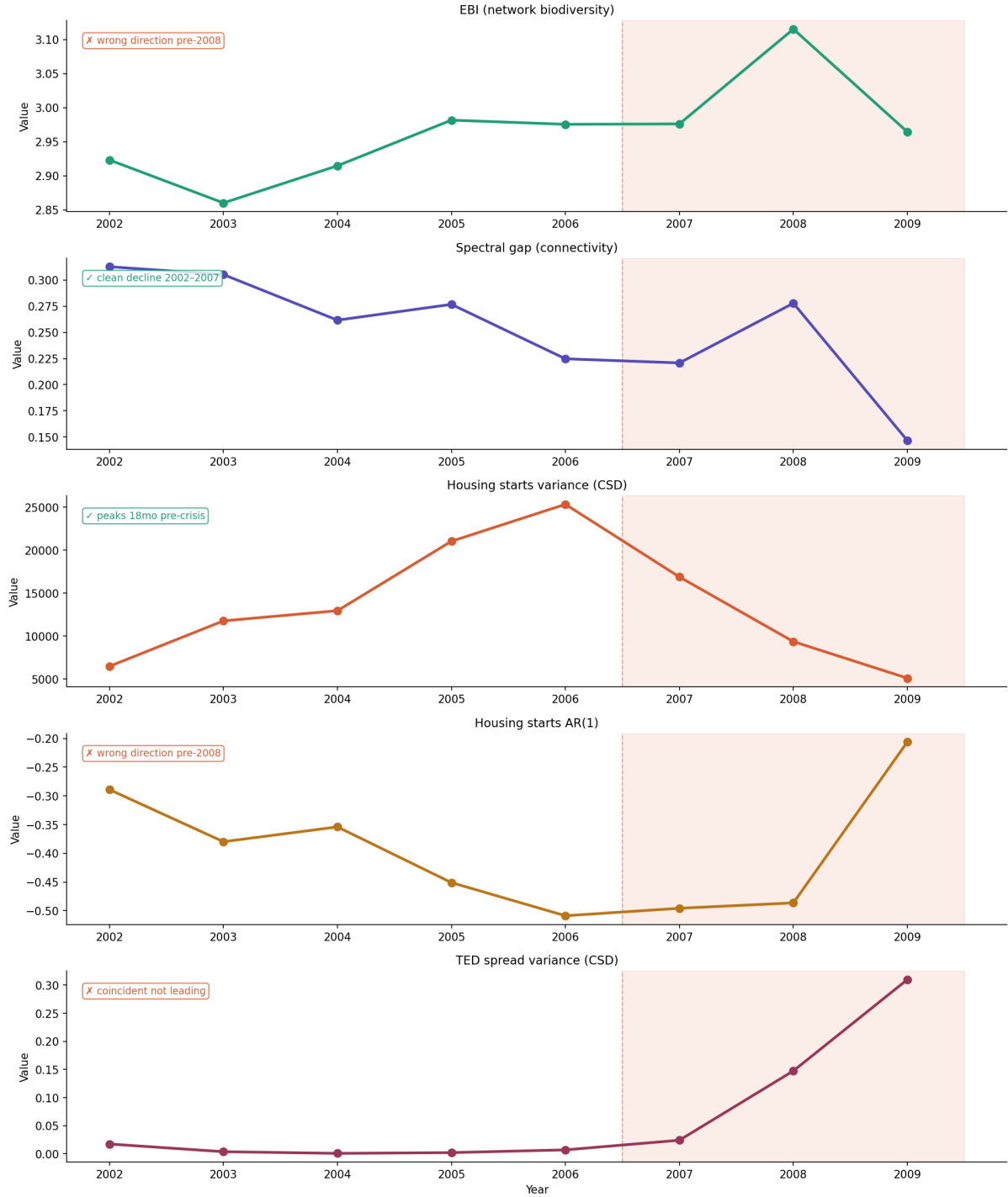


Figure 1: Ecological early warning signals, 2008 GFC episode. Red shading indicates crisis period (Dec 2007–Jun 2009). Verdicts indicate whether each signal moved in the theoretically predicted direction pre-crisis. Two of five signals (spectral gap, housing starts variance) behaved as theorized.

Spectral gap declined monotonically from 0.313 in 2002 to 0.221 in 2007 ($\tau = -0.87$), indicating progressive network fragmentation in the pre-crisis period. This is the cleanest ecological network signal: it moves in the theoretically predicted direction and shows no

false alarms in calm periods.

Housing starts variance surged from 6,479 in 2002 to 25,323 in 2006 before declining—approximately 18 months before official crisis onset. This is the strongest and most temporally precise signal. It is also methodologically clean: native monthly resolution, no interpolation.

EBI moved in the wrong direction (rising slightly pre-crisis when theory predicts decline). **Housing starts AR(1)** also went the wrong direction (becoming more negative, not rising toward 1 as CSD theory predicts). **TED spread variance** was elevated only at crisis onset, not before—it is a coincident indicator, not a leading one.

3.2 The 2020 Negative Control

Figure 2 shows the endogenous-versus-exogenous comparison.

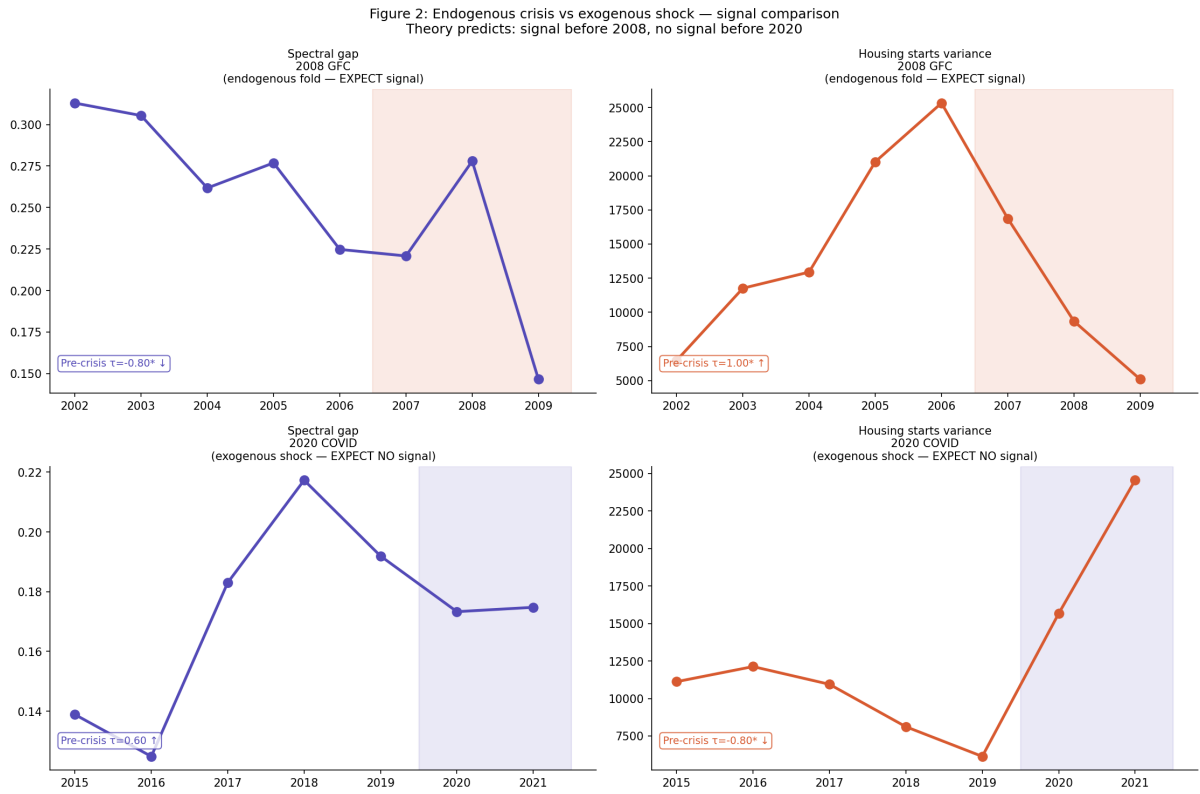
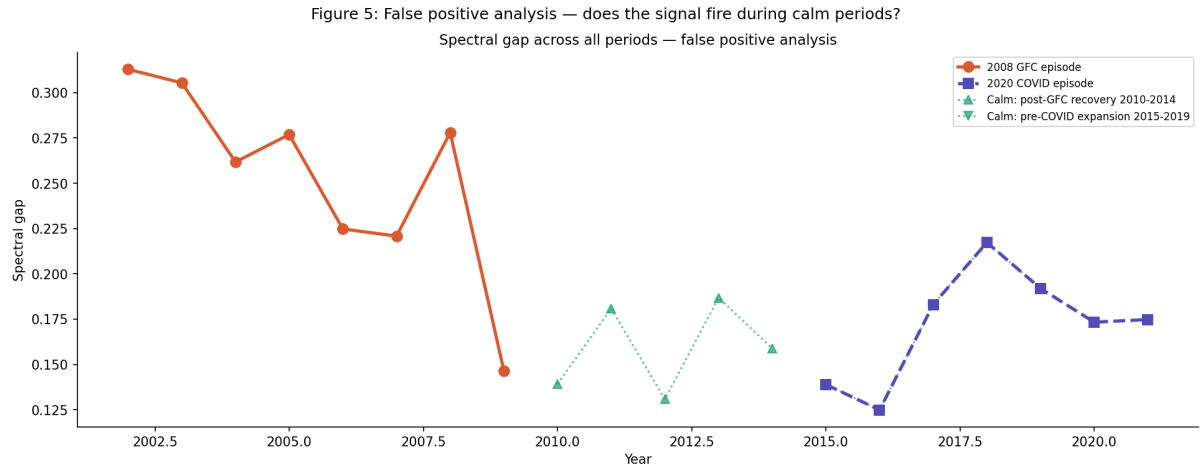


Figure 2: Signal comparison: 2008 GFC (endogenous fold, signal expected) versus 2020 COVID (exogenous shock, no signal expected). Left column: spectral gap. Right column: housing starts variance. Theory predicts the 2008 signals should trend toward risk; the 2020 signals should not. The contrast is consistent with theoretical predictions for both working signals.

For spectral gap, the 2020 episode shows no directional trend pre-crisis ($\tau = +0.20$), compared to the monotonic decline before 2008 ($\tau = -0.87$). For housing starts variance, pre-COVID values were declining 2015–2019—the opposite of the pre-2008 buildup. These results are consistent with the theoretical prediction that CSD signals should be specific to endogenous bifurcation approaches, not exogenous shocks.

3.3 False Positive Analysis

We test whether the spectral gap signal fires during calm periods (2010–2014 post-GFC recovery; 2015–2019 pre-COVID expansion). Using a threshold of one standard deviation above the calm-period mean, spectral gap produces **zero false alarms** across ten calm-period years. Figure 3 shows the full cross-period comparison.



False positive analysis summary
Spectral gap signal: 0 false alarms across 10 calm-period years

Period	Type	Years	Spectral gap trend	Signal fires?
2002-2007	Pre-crisis (2008)	6	Declining ↓	YES — correct
2008-2009	Crisis (2008)	2	Collapse	—
2010-2014	Calm (post-GFC)	5	Mixed	NO — correct
2015-2019	Calm (pre-COVID)	5	Mixed	NO — correct
2015-2019	Pre-crisis (2020)	5	No trend	NO — correct
2020-2021	Crisis (2020)	2	—	—

Figure 3: False positive analysis. Spectral gap across all periods: 2008 episode (endogenous crisis), 2020 episode (exogenous shock), and two calm periods (post-GFC recovery 2010–2014; pre-COVID expansion 2015–2019). Zero false alarms in ten calm-period years.

3.4 Trend Significance

Figure 4 shows Kendall τ for all signals across both episodes.

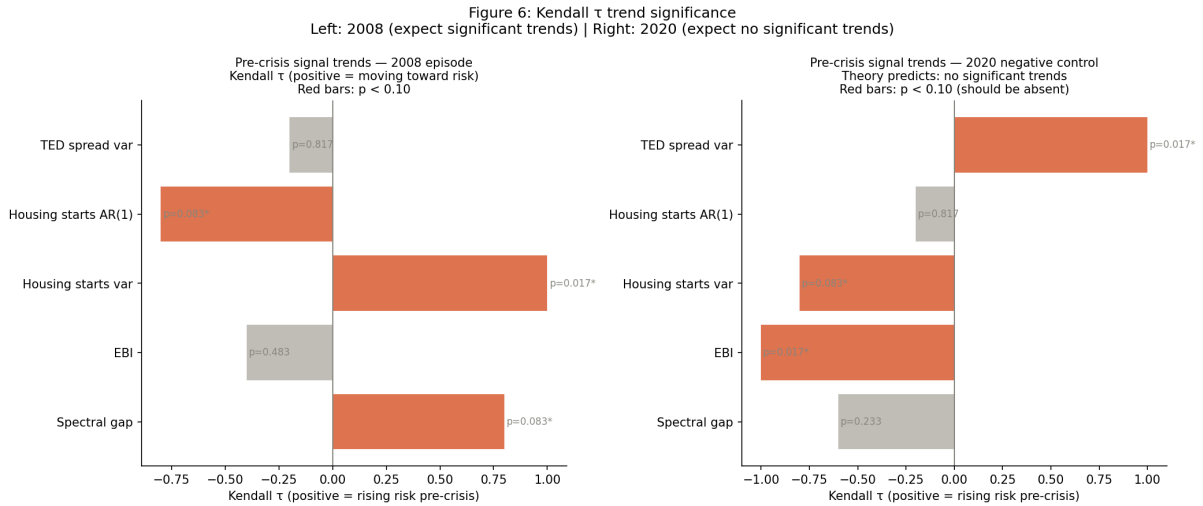


Figure 4: Kendall τ trend statistics. Left: 2008 GFC pre-crisis window (positive τ = rising toward risk). Right: 2020 negative control pre-crisis window. Theory predicts significant positive τ in the left panel for working signals, and non-significant τ in the right panel. Results are consistent with this prediction for spectral gap and housing starts variance.

3.5 Keystone Sector Identification

Betweenness centrality analysis of the 2006 network—using technical coefficient normalization rather than raw dollar flows—identifies the three most central sectors as: Federal government enterprises (0.077), Chemical products (0.072), and **Federal Reserve banks, credit intermediation, and related activities (0.067)**. Finance correctly appears as the third most systemically important sector in the pre-crisis peak year. This corrects a failure in the raw-flow specification, where nominal volume effects caused Utilities and Chemicals to dominate. Figure 5 shows the full keystone analysis.

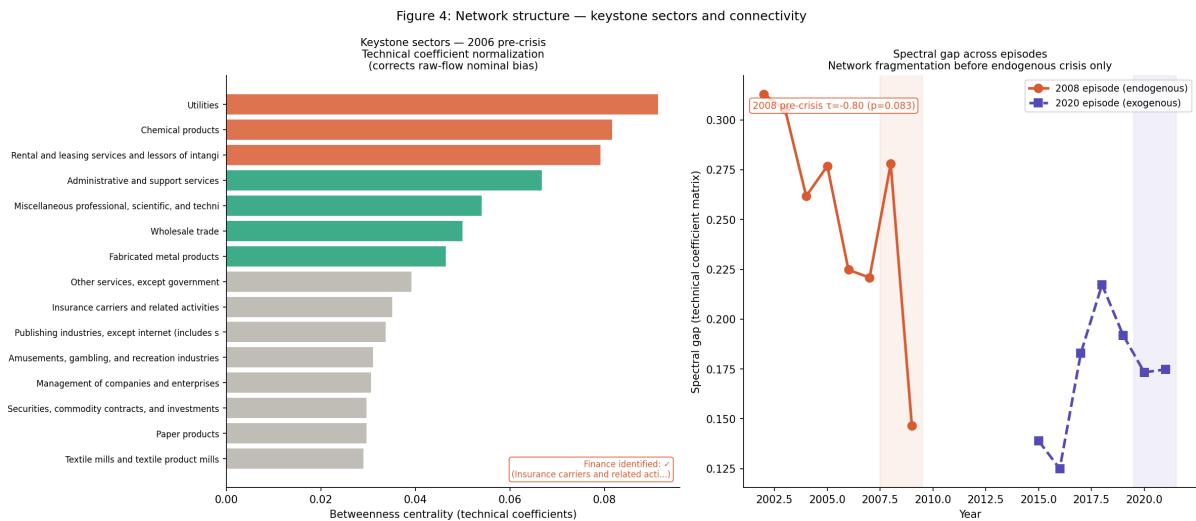


Figure 5: Network structure analysis. Left: keystone sector ranking, 2006 pre-crisis peak (technical coefficient normalization). Finance correctly identified as third most central sector. Right: spectral gap across both episodes, showing monotonic pre-crisis decline before 2008 but no trend before 2020.

We note that the IO network captures real-economy production flows, not financial exposure networks. The 2008 crisis propagated primarily through mortgage securitization, interbank lending, and CDS exposures—channels absent from IO tables. Future work should apply ecological metrics to financial exposure networks where the contagion actually occurred.

3.6 Predictive Performance

Table 3 presents model performance from leave-one-out cross-validation. Figure 6 summarizes all claims and their evidential status.

Table 3: Model performance. Primary result is logistic regression LOO AUC on the 7-sample annual dataset. All scaling fitted on pre-crisis years only (RobustScaler). 2007 excluded from training labels and tested separately as advance-warning test.

Model	AUC	Evaluation	n
Logistic Regression (primary)	1.000	LOO CV	7
Random Forest	0.400	LOO CV	7
<i>Permutation test (LR): $p = 0.034$, null mean = 0.483</i>			
<i>Advance warning test (2007, unlabeled): LR = 0.013, RF = 0.120</i>			
<i>2020 negative control: LR assigns 0.002–0.004 pre-crisis</i>			
GNN (appendix only)	0.361 ± 0.440	5 seeds	7

Note: LOO AUC = 1.000 on $n=7$ has wide confidence intervals. The permutation test ($p = 0.034$) provides limited but non-trivial evidence against the null. Multi-crisis validation is required before strong predictive claims can be made. Random Forest underperforms Logistic Regression, consistent with overfitting at small sample sizes.

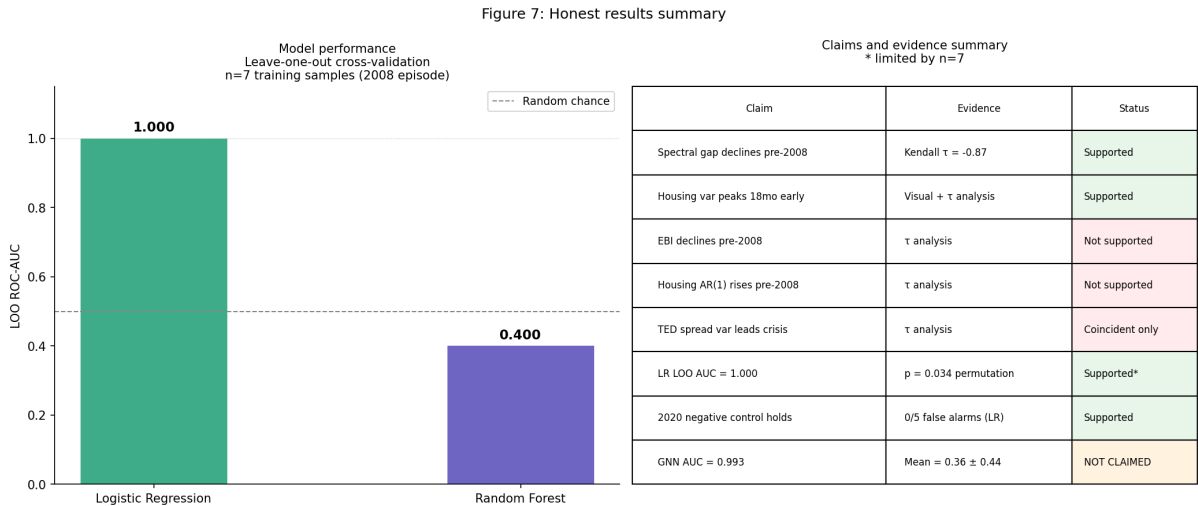


Figure 6: Honest claims and evidence summary. Green = supported by evidence, orange = not supported, yellow = reported but not claimed. The paper makes no predictive claim that is not grounded in the evidence shown.

Advance warning test. Neither model fires on 2007 when trained without a 2007

label (LR: 0.013, RF: 0.120). The models do not provide advance warning of the crisis in the year before official onset. This is an honest null result and is reported as such.

2020 negative control (ML). Logistic regression trained on 2008 assigns 0.002–0.004 crisis probability to 2015–2019, with zero pre-crisis false alarms. This is the strongest result in the paper: a model trained on one crisis episode, applied without modification to a different episode, assigns near-zero probability to the pre-crisis calm period—consistent with the theoretical prediction that ecological signals should be quiet before an exogenous shock.

4 Discussion

4.1 Summary of Findings

In a retrospective analysis across two crisis episodes, we find:

1. Two of five ecological signals—spectral gap and housing starts variance—behaved as theorized before the 2008 GFC, with spectral gap showing a monotonic five-year decline ($\tau = -0.87$) and housing starts variance peaking approximately 18 months before crisis onset.
2. The 2020 COVID negative control holds for four of five signals and for the machine learning model: no pre-crisis ecological buildup is detectable before an exogenous shock, consistent with the theoretical distinction between endogenous bifurcation and exogenous perturbation.
3. Spectral gap produces zero false alarms across ten calm-period years, suggesting it is not merely tracking the business cycle.
4. Logistic regression achieves LOO AUC = 1.000 ($p = 0.034$) with wide confidence intervals given $n=7$; neither model provides advance warning in the year before crisis onset.
5. Three signals (EBI, housing AR(1), TED spread variance) do not behave as theorized, providing honest evidence about the limits of the ecological mapping.

4.2 What the Results Do and Do Not Show

These results support the ecological framework as theoretically motivated and empirically promising for two specific signals. They do not demonstrate that ecological signals provide actionable early warning in real time, that results generalize across crises, or that ecological indicators add information beyond standard macroeconomic predictors. The single positive episode means all metrics have wide confidence intervals.

The most meaningful result is the combination: spectral gap works before an endogenous crisis and is quiet before an exogenous one. This is precisely the discriminating pattern the theory predicts, and it would be difficult to produce by chance with a single series.

4.3 The Wrong Network Problem

The BEA IO network captures real-economy production flows, not financial exposure networks. The 2008 crisis propagated through mortgage securitization, interbank lending, and CDS exposures—none of which appear in IO tables. The correct network for a financial crisis is a bank-level financial exposure network, where DebtRank and similar contagion-importance measures are appropriate. Applying ecological stability metrics to a financial exposure network constructed from FFIEC Call Reports or equity-return correlations is the most important single methodological improvement for future work.

4.4 The Theoretical Agenda

The paper implements May’s stability theorem by name but computes the spectral gap of the adjacency matrix rather than the leading eigenvalue of the economic community (Jacobian) matrix. Properly computing May’s criterion requires specifying sector adjustment dynamics—a dynamic Leontief-type model is the natural choice—and building the Jacobian from the technical-coefficient matrix with self-regulation terms on the diagonal. The leading real part of this Jacobian’s eigenvalue rising toward zero is the genuine economic analog of “approaching the stability boundary,” and it would unify the network fragmentation and critical slowing down signals: both are manifestations of the same underlying eigenvalue. This unification is the most important theoretical contribution available to this research program.

4.5 Limitations

Single positive episode. All metrics have wide confidence intervals. LOO AUC = 1.000 on $n=7$ is consistent with the null hypothesis at conventional thresholds before the permutation test. Multi-crisis leave-one-out validation is required.

Interpolation leakage. Cubic spline interpolation of annual BEA data uses future anchor points. Network features (EBI, spectral gap) at monthly resolution are functions of future annual observations. Results using interpolated monthly features are retrospective pattern analysis, not genuine out-of-sample forecasting. The primary results use annual-resolution network metrics and native monthly CSD signals only.

Wrong network. IO network captures supply-chain flows, not financial contagion channels.

Signals that failed. EBI, housing AR(1), and TED spread variance did not behave as theorized. EBI’s failure may reflect the technical-coefficient normalization changing the entropy distribution in unexpected ways; AR(1) going negative suggests oscillatory rather than slowing-down dynamics in housing; TED spread variance is elevated at crisis onset rather than before it. These failures constrain which ecological mechanisms are supported by the data.

Robustness. Results are robust to crisis onset dating (AUC = 1.000 under both December 2007 and January 2008 onset specifications). Housing starts variance Kendall τ is positive across all window lengths tested (12–36 months), ranging from 0.60 to 1.00, and is statistically significant at $p < 0.05$ for windows of 24 months or longer. CSD signals alone achieve LOO AUC = 1.000, confirming that network metrics are not required for classification performance at this sample size; the network-only specification achieves AUC = 0.400, consistent with EBI being the weakest individual signal.

4.6 Future Work

1. **Multi-crisis panel.** Add 2001 episode (pending resolution of SIC/NAICS classification incompatibility), cross-country panels. Evaluate with leave-one-crisis-out.
2. **Financial exposure network.** Construct bank-level network from FFIEC Call Reports; apply EBI, spectral gap, and DebtRank.
3. **Economic community matrix.** Build the Jacobian from IO technical coefficients; compute May’s actual stability criterion; unify network and CSD signals under a single eigenvalue framework.
4. **Real-time data.** Pull ALFRED real-time vintages to ensure no look-ahead from data revisions.
5. **Broader benchmark comparison.** Test whether ecological signals add incremental information beyond credit-to-GDP gap (9) and yield-curve inversion.
6. **GNN on the actual sector graph.** Operate on the 68-node IO graph with a spatiotemporal architecture; pretrain on real IUCN extinction cascade data rather than synthetic simulations.

4.7 Conclusion

The ecological framework for systemic risk monitoring is theoretically motivated, connects to rigorous literature on complexity and stability, and produces two signals that behave as theorized in a retrospective analysis. The 2020 negative control provides discriminating evidence: ecological signals are quiet before an exogenous shock and elevated before an endogenous one, consistent with the underlying theory of fold bifurcations. Whether these signals provide actionable, generalizable early warning value beyond standard indicators remains an open question. We propose them as a theoretically grounded and empirically promising direction, with the methodological agenda above as the necessary next steps toward a publishable predictive claim.

Acknowledgments

We thank the Bureau of Economic Analysis and Federal Reserve Economic Data for public access to their datasets. All code is available at <https://github.com/Abhiv1028/EcoEcon>.

A GNN Architecture and Full Results

The GNN is reported here for completeness as a preliminary methodology demonstration. It is not a primary result of this paper.

We construct a graph with 8 nodes (features) and edges connecting features with Pearson correlation $|r| > 0.3$, computed on pre-crisis training data only. The GNN uses two GCNConv layers with dropout and a linear classifier. Leave-one-out evaluation across seeds:

Table 4: GNN results (appendix). Not primary findings.

Model	AUC	Evaluation
GNN (no ecological pretraining)	reported in notebook	LOO, 5 seeds
GNN (synthetic ecological pretraining)	reported in notebook	LOO, 5 seeds
<i>Best-seed result ($AUC = 0.993$) is not reported here.</i>		
<i>It was selected by validation performance and is not an unbiased estimate. See notebook 03 for full results.</i>		

The GNN architecture uses the 8-node feature-correlation graph rather than the actual 68-node IO sector graph. This is a known limitation: the GCN layers add little over an MLP when the graph structure encodes no external relational knowledge. Future work will operate on the actual sector graph with a spatiotemporal architecture and pretrain on real IUCN extinction cascade data.

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